◀ FOLD 2: FOLD RIGHT BEHIND LEFT (01 ON COVER) | UNFOLD STEP 1: OPEN RIGHT ▶

UNIT CIRCLE VOLVELLE & TRIGONOMETRIC CALCULATOR

Spin-to-Read Rotating Wheel Chart for All 16 Special Trig Angles

WHAT IS THIS BOX FOR? NEVER MEMORIZE A TRIG TABLE AGAIN.

Sines and cosines aren't random decimals to cram before a test—they're just the X and Y shadows of a spinning wheel of radius 1.

WELCOME TO THE POCKET TRIG WHEEL

Instead of memorizing 16 separate angles, spin one dial:

- **Degrees & Radians Together: See 30° and π/6 lined up side-by-side**
- **Exact Radical Windows: Read (cos θ, sin θ) as (√3/2, 1/2) instantly**
- **Rotating Radius Arm: Sweeps a unit radius (r = 1.00) around all 4 quadrants**
- Automatic Sign Check: Shows where X and Y turn negative in Q2, Q3, and Q4

Try this right now: Turn the pointer to 45° (π/4) and read the window!

WHY IS IT CALLED A "VOLVELLE"?

Medieval astronomers used rotating paper and brass wheel charts—called volvelles—to compute star positions without pen and paper. Yours computes trig!

WHAT'S IN YOUR BOX (5 PIECES)

- 1x Base Stator Disc (0°–360° Special Angles & X/Y Coordinate Grid)
- 1x Rotating Radian Dial (Gold) + 1x Radial Cursor Arm
- 1x Snap-Fit Center Hub Pin + 1x Protective Travel Slipcase

★ YOUR FIRST 60 SECONDS WITH THIS BOX

START HERE!

1. DIAL THE CURSOR ARM TO 30° (π/6):
Look through the cursor window: read Degrees (30°), Radians (π/6), and Exact Coordinates (√3/2, 1/2)!

2. COMPARE 30° AND 60° (π/3):
Spin to 60°: notice how √3/2 and 1/2 simply swap places because the triangle flips across 45°!

3. PROJECT WITH [↱][▸y,x] ON SC-32:
Type 30 [ENTER] 1 [↱][▸y,x] on your SC-32 to get X = 0.8660 (√3/2) and Y = 0.5000 (1/2) simultaneously!

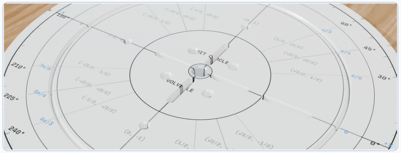
DISC ICON = CALCULATOR [COS], [SIN] & [↱][▸Y,X] PHASOR KEYS

The button icon embossed on your box lid (left) is the exact calculator key ([DEG▸RAD] & [↱][▸y,x])! Build every lesson with your blocks, then press the key.

EXACT RADICALS (√3/2) ↔ [COS] & [√x] VERIFICATION

STEP A: HANDS-ON WITH THE VOLVELLE DISC

1:1 CAD SCENE



cos(30°) = √3/2 ≈ 0.8660 on SC-32	cos(45°) = √2/2 ≈ 0.7071 on SC-32	sin(30°) = 1/2 = 0.5000 on SC-32

▲ **Read the exact symbolic radical √3/2 in your Volvelle window,** then verify 3 [√x] 2 [÷] = 30 [COS] on SC-32!

STEP B: TRY IT ON YOUR CALCULATOR

BOX ICON: [DEG▸RAD] & [↱][▸y

RADICAL TREE: θ = 30° (π/6) RIGHT TRIANGLE

θ = 30° (π/6 RAD)
Volvelle: √3/2
SC-32: 30 [COS]
3 [√x] 2 [÷]
Y=0.8660
X=0.8660
3 [√x] 2 [÷] 30 [COS] → Y=X=0.8660

PRESS BOX ICON & RPN KEYS (SC32):

1. RADICAL √3/2:
3 √x 2 ÷ 0.8660

2. COMPARE COS:
30 COS Y=X=0.8660

✓ DISPLAY (X) = 0.8660254038

KEYSTROKES: 3 [√x] 2 [÷] 30 [COS]
Evaluate √3/2 from the Volvelle window with 3 [√x] 2 [÷], then press 30 [COS] to see Y and X match!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

HOW IT WORKS: THE CORE IDEA

Unit Circle Coordinates: (cos θ, sin θ) on a Circle of Radius 1

HOW THE UNIT CIRCLE CONNECTS GEOMETRY & TRIG

A unit circle is simply a circle of radius 1 centered at (0, 0): $x^2 + y^2 = 1$.
When you turn the pointer arm to any angle θ from the positive x-axis:

Horizontal Shadow $x = \cos(\theta)$ | Vertical Height $y = \sin(\theta)$

Because the pointer arm always has length 1, the right triangle it forms with the x-axis automatically obeys Pythagoras:

$$\cos^2(\theta) + \sin^2(\theta) = 1^2 = 1$$

And the slope (steepness) of the pointer arm is the tangent:

$$\tan(\theta) = \text{Rise} / \text{Run} = \sin(\theta) / \cos(\theta)$$

ALL 16 SPECIAL ANGLES AT YOUR FINGERTIPS

Every multiple of 30° (π/6) and 45° (π/4) around the circle comes from just two classic right triangles: the 30–60–90 triangle and the 45–45–90 square diagonal. Once you know Quadrant I, the other three quadrants are mirror images!

NEVER MISS A MINUS SIGN AGAIN

Left of center? Cosine (x) is negative. Below center? Sine (y) is negative. Looking at the physical pointer position makes plus and minus signs obvious at a glance, without relying on fragile mnemonics.

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE

VISUAL PROOF

✗ **TRAP 1 – Confusing which is sin and which is cos:**
✓ **FIX:** Alphabetical order! C comes before S, just like X comes before Y: (cos θ, sin θ) = (x, y).

✗ **TRAP 2 – Mixing up √3/2 and 1/2 at 30° vs 60°:**
✓ **FIX:** Look at the physical wheel: at 30°, the point is wide (x = √3/2 ≈ 0.87) and low (y = 1/2 = 0.50)!

✗ **TRAP 3 – Reversing θ and r in [↱][▸y,x]:**
✓ **FIX:** On SC-32, push angle θ into Y first, type radius r in X, then press [↱][▸y,x]: 30 [ENTER] 1 [↱][▸y,x]!

CALCULATOR CHECK: 3 [√x] 2 [÷] 30 [COS] → X = 0.8660254038

SPECIAL ANGLES & EXACT RADICAL VALUES

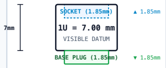
DEGREES	RADIANS	COS (X)	SIN (Y)	TAN
0°	0	1	0	0
30°	π / 6	√3 / 2	1 / 2	√3 / 3
45°	π / 4	√2 / 2	√2 / 2	1
60°	π / 3	1 / 2	√3 / 2	√3
90°	π / 2	0	1	undef
120°	2π / 3	−1 / 2	√3 / 2	−√3
180°	π	−1	0	0
270°	3π / 2	0	−1	undef

DESIGNED FOR DESK & BACKPACK

- Compact 110 mm outer disc slips easily into a binder pouch or jacket pocket.
- Smooth snap-fit center hub holds your chosen angle steady while you read.
- Deeply embossed angle ticks and radical fractions never fade or rub off.
- Pairs directly with the SC-32 calculator's [SIN], [COS], and [TAN] keys.

1:1 CAD REFERENCE VIEW

1:1 BLOCK CAD ANATOMY



✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely:
Only 7.00mm true heights add!

BOX ICON & FORMULA QUICK-REFERENCE

Box / Disc Icon Keys:
[COS], [SIN], [↱][▸RAD] & [↱][▸y,x]

Coordinate Law:
(x, y) = (cos θ, sin θ) | tan θ = sin θ / cos θ

Pythagorean Identity:
$\cos^2(\theta) + \sin^2(\theta) = 1^2 = 1$ (for ALL θ)

30° (π/6) ↔ 60° (π/3):
30°: (√3/2, 1/2) | 60°: (1/2, √3/2)

SC-32 Polar → Rect:
θ [ENTER] r [↱] [▸y,x] → X=r·cosθ, Y=r·sinθ

GEOMETRY MEETS TRIGONOMETRY • COSINE IS X, SINE IS Y

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

EXAMPLE 1: HANDS-ON WALKTHROUGH

Dial In 45° (π/4): Read Exact Radicals (√2/2, √2/2) in the Window

① ALIGNING SPECIAL ANGLES & RADICALS

1. ROTATE INNER ROTOR TO 45° (π/4):
Grip the textured rotor rim and rotate until the cursor line aligns exactly with the 45° (π/4) graduation tick.

2. READ EXACT VALUES THROUGH THE CURSOR WINDOW:

- Radial Position: Exact unit circle boundary (r = 1.00)
- **Horizontal Cosine (x): cos(45°) = √2 / 2 ≈ 0.7071**
- **Vertical Sine (y): sin(45°) = √2 / 2 ≈ 0.7071**
- **Tangent Slope: tan(45°) = 1.000 (Equal legs: x = y)**

3. TEST OTHER SPECIAL ANGLES:
Rotate to 30° (π/6): cos = √3/2, sin = 1/2. Rotate to 60° (π/3): cos = 1/2, sin = √3/2.

ELIMINATE PENCIL & PAPER GUESSWORK

Students frequently swap √3/2 and 1/2 on exams. Looking at the volvelle makes it visually undeniable: at 30°, the horizontal base is much longer than the height, so cos(30°) must be √3/2 (0.866) and sin(30°) must be 1/2 (0.5)!

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR

[DEG▸RAD] & [↱][▸y

STEP 1 (DIAL 45° / π/4):
Align the Volvelle cursor arm with 45° → window reads (√2/2, √2/2) and tan(45°) = 1.

STEP 2 (SQUARE COS 45°):
On your SC-32 calculator, press 45 [COS] [↰] [x^2] → X = 0.5000.

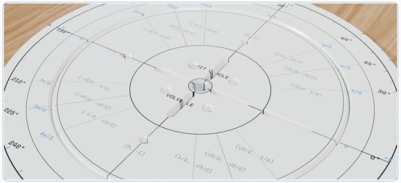
STEP 3 (ADD SIN^2 45°):
Press 45 [SIN] [↰] [x^2] [+] → 0.5 + 0.5 = 1 (proving $\cos^2\theta + \sin^2\theta = 1$)!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 1 → DISPLAY (X) = 1

LESSON 1: DIAL 45° (π/4) ON VOLVELLE ↔ PYTHAGOREAN KEYST

STEP A: HANDS-ON WITH THE VOLVELLE DISC

1:1 CAD SCENE



16 Special Angles Degrees (°) & Radians (	Exact Radicals (cos θ, sin θ) in Windo	Radius R = 1 $\cos^2\theta + \sin^2\theta = 1$

▲ **Dial the cursor arm to 45° (π/4): the window frames (√2/2,** √2/2), and squaring + adding on SC-32 gives 1!

STEP B: TRY IT ON YOUR CALCULATOR

BOX ICON: [DEG▸RAD] & [↱][▸y

PYTHAGOREAN TREE: θ = 45° (π/4 RAD)

$\cos^2(45°) + \sin^2(45°) = 1$
45 [COS] [↰][x^2]
45 [SIN] [↰][x^2]
0.7071² = 0.5
0.7071² = 0.5
[+] = 1
45 [COS][↰][x^2] 45 [SIN][↰][x^2] [+] → 1

PRESS BOX ICON & RPN KEYS (SC32):

1. COS^2(45°):
45 COS ↰ x^2 = 0.5

2. + SIN^2(45°):
45 SIN ↰ x^2 + = 1

✓ DISPLAY (X) = 1

KEYSTROKES: 45 [COS] [↰] [x^2] 45 [SIN] [↰] [x^2] [+]
Dial 45° on the Volvelle, then square cos(45°) and sin(45°) on SC-32 (no redundant ENTER before [↰][x^2])!
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 1 → X = 1

EXACT RADICAL FRACTIONS • NEVER CONFUSE √3/2 AND 1/2 AGAIN

CONTINUE RIGHT → PAGE 04

EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

Pythagorean Identity: Why $\cos^2(\theta) + \sin^2(\theta)$ Always Equals 1

② PYTHAGOREAN IDENTITY: $\cos^2\theta + \sin^2\theta = 1$

THE BIG DISCOVERY: Why does $\cos^2\theta + \sin^2\theta$ ALWAYS equal 1?

1. TRACE LEGS AT ANY ANGLE θ:
- The horizontal base of the triangle is length |cos θ|.
- The vertical height of the triangle is length |sin θ|.
- The hypotenuse is the radial arm connecting origin to (cos θ, sin θ).

2. APPLY THE PYTHAGOREAN THEOREM:

$$\text{Base}^2 + \text{Height}^2 = \text{Hypotenuse}^2$$
$$(\cos\theta)^2 + (\sin\theta)^2 = (1.00)^2 = 1$$

3. ROTATE PAST 90° INTO QUADRANT II (e.g. 120°):
cos(120°) = −1/2, sin(120°) = √3/2.

$(-1/2)^2 + (\sqrt{3}/2)^2 = 1/4 + 3/4 = 4/4 = 1.00$! **(Signs square away!)**

✓ UNIT RADIUS CONSERVATION THEOREM

The identity is not a separate formula to memorize.
It is literally the Pythagorean theorem on the circle!
No matter how fast or far you spin the volvelle rotor, the tip of the radial arm always remains exactly 1 unit from the origin.
Pythagoras and trigonometry are one single unified truth.

★ 2-PLAYER GAME: 'QUADRANT RADICAL SPRINT' (2 MINS)

Player A calls out a radian angle (like 5π/6 or 7π/4). Player B has 5 seconds to predict the plus/minus signs of (cos, sin) and the reference triangle (30°, 45°, or 60°), spins the **Volvelle cursor arm to verify, and checks with θ [ENTER] 1 [↱][▸y,x] on SC-32!**

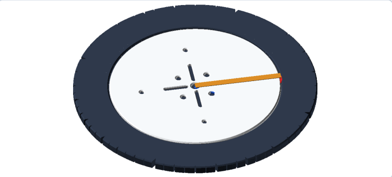
⚙ WHERE THIS SHOWS UP IN AUDIO SYNTHESIS, AC POWER & GAME ENGINES

Why are 'volvelles' shaped like rotating discs? Astronomers used paper and brass volvelles for centuries before electronic computers! Every 3D video game rotation matrix, musical synthesizer **oscillator, and AC electrical grid wave runs on (cos ωt, sin ωt).**

LESSON 2: 30° VOLVELLE WINDOW ↔ [↱][▸y,x] POLAR-TO-RECT

STEP A: HANDS-ON WITH 30° (π/6) PROJECTION

1:1 CAD SCENE



30° = π/6 rad (√3/2, 1/2)	X = cos 30° 0.8660 on SC-32	Y = sin 30° 0.5000 on SC-32

▲ **Dial 30° on the Volvelle, then press 30 [ENTER] 1 [↱][▸y,x] on** SC-32 to project X=0.8660 and Y=0.5000!

STEP B: TRY IT ON YOUR CALCULATOR

BOX ICON: [DEG▸RAD] & [↱][▸y

POLAR-TO-RECT PROJECTION: θ=30°, r=1

30° [ENTER] 1 [↱][▸y,x]
X = r·cos(30°)
Y = r·sin(30°)
X = 0.8660
(√3/2)
Y = 0.5000
30 [ENTER] 1 [↱][▸y,x] → X=0.8660, Y=0.5

PRESS BOX ICON & RPN KEYS (SC32):

1. θ IN Y, r IN X:
30 ENTER 1 ↱ ▸y,x

2. READ X & Y:
X=0.8660 Y=0.5000

✓ DISPLAY (X) = 0.8660254038

KEYSTROKES: 30 [ENTER] 1 [↱] [▸y,x]
Enter angle θ = 30° into Y and radius r = 1 into X, then press [↱][▸y,x] to project both coordinates at once!
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0.5 | X: 0.8660254038 → X = 0.8660254038

PYTHAGORAS ON A CIRCLE • HYPOTENUSE IS ALWAYS EQUAL TO 1

STEP 3: FLIP SHEET OVER ↻ PAGE 05

EXAMPLE 3 & DESK CHALLENGES

Reference Angles Across All 4 Quadrants + 3 Desk Challenges

③ SYMMETRY ACROSS QUADRANTS: 30°, 150°, 210°, AND 330°

THE GOAL: See why four different angles share the exact same (√3/2, 1/2) numbers!

1. START AT 30° (π/6) IN QUADRANT I:
Read the window: cos(30°) = +√3/2 (long horizontal), sin(30°) = +1/2 (short height).

2. SPIN TO 150° (5π/6) IN QUADRANT II:
It's the exact same 30° triangle flipped left! Now x is negative: (−√3/2, +1/2).

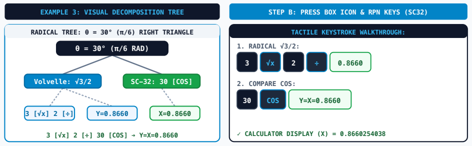
3. CHECK 210° (QIII) AND 330° (QIV):
At 210° both are negative (−√3/2, −1/2); at 330° only y is negative (+√3/2, −1/2).

✓ HOW 3 TRIANGLES UNLOCK ALL 16 SPECIAL ANGLES

You don't need to memorize 16 separate coordinate pairs around the circle.
Every angle is just a 30°, 45°, or 60° right triangle reflected across the X and Y axes!
Look at the pointer arm first: is it left of center (cos < 0)? Below center (sin < 0)?
Verify any angle on your SC-32 calculator: press [3][0][COS] to see 0.8660 (√3/2)!

PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!


EXAMPLE 3: VISUAL DECOMPOSITION TREE
RADICAL TREE: θ = 30° (π/6) RIGHT TRIANGLE
θ = 30° (π/6 RAD)
Volvelle: √3/2
SC-32: 30 [COS]
3 [√x] 2 [÷]
Y=0.8660
X=0.8660
3 [√x] 2 [÷] 30 [COS] → Y=X=0.8660
STEP B: PRESS BOX ICON & RPN KEYS (SC32)
TACTILE KEYSTROKE WALKTHROUGH:
1. RADICAL √3/2:
3
√x
2
÷
0.8660
2. COMPARE COS:
30
COS
Y=X=0.8660
✓ CALCULATOR DISPLAY (X) = 0.8660254038

FULL KEYSTROKES: 3 [√x] 2 [÷] 30 [COS]
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0.8660254038 | X: 0.8660254038 → X = 0.8660254038

3 DESK CHALLENGES — TRY THESE NOW!

WARM-UP #1 **Turn to 150° (5π/6). What are the exact values of cos θ and sin θ?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ **Build first, then flip booklet upside-down to check Answer #1 below!**

LEVEL-UP #2 **Turn to 225° (5π/4) in Quadrant III. Why is tan(225°) positive +1?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ **Build first, then flip booklet upside-down to check Answer #2 below!**

BRAIN-MELTER #3 **At which two angles around the circle is tan(θ) undefined?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ **Build first, then flip booklet upside-down to check Answer #3 below!**

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)
#3: 90° (π/2) and 270° (3π/2), where x = 0
#1: cos = −√3/2, sin = +1/2 | #2: (−√2/2) ÷ (−√2/2) = +1!

FOR PARENTS & TEACHERS (Grades 9–12 & College (Geometry, Pre-Calculus & Calculus))

- CCSS.MATH.CONTENT.HSF.TF.A.2: Explain how the unit circle extends trig functions to all real numbers.
- CCSS.MATH.CONTENT.HSF.TF.A.3: Use special triangles to determine geometrically the values of sin, cos, tan.
- CCSS.MATH.CONTENT.HSF.TF.C.8: Prove the Pythagorean identity $\sin^2(\theta) + \cos^2(\theta) = 1$.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

BOSS CHALLENGE #4 **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([DEG▸RAD] & [↱][▸y,x]) and keystrokes on your calculator:

- **3 [√x] 2 [÷] 30 [COS] → X = 0.8660254038**

CHECK OFF YOUR MASTERY MILESTONES:

- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ **4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)**

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

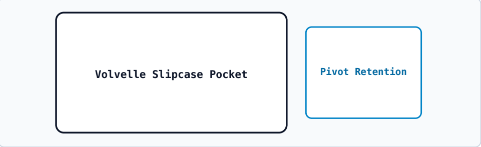
NEXT: HOW TO PACK UP → PAGE 06

HOW TO STACK IT ALL BACK IN

The 10-Second Self-Checking Pack-Up Puzzle

TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

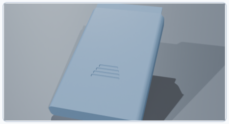
Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:



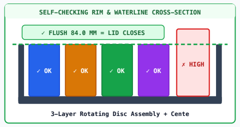
✓ SLIDES ASSEMBLED INTO THE PROTECTIVE SLIPCASE

You never need to take the volvelle apart after assembly—the base disc, gold radian rotor, cursor arm, and center pin stay snapped together as one pocket instrument. When you finish your problem set, align the cursor arm with 0° and slide the entire wheel straight into its protective travel slipcase!

1. STORAGE CASE & COMPARTMENTS



2. WHY THE LID SELF-CHECKS EVERY PIECE


SELF-CHECKING RIM & WATERLINE CROSS-SECTION
✓ FLUSH 84.0 MM = LID CLOSES
✓ OK
✓ OK
✓ OK
✓ OK
✗ HIGH
3-Layer Rotating Disc Assembly + Cente

▲ SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

THE 10-SECOND PACK-UP ROUTINE

HOW TO STACK EVERYTHING BACK IN:

1. SORT BY COLOR & NUMBER:
Group matching blocks or tiles together on your desk first.

2. STACK & DROP INTO LABELED BAYS:
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!

3. TUCK THIS CARD IN THE LID & SNAP SHUT:
Fold this guide back to [01] and rest it right on top before closing.

HUB PIN CARE & BINDER STORAGE

- The snap-fit center pin holds your chosen angle steady without loosening over time.
- If dust ever gets under the rotor, rinse the wheel in cool water and pat dry.
- Tuck this folded reference guide right inside the slipcase pocket alongside the wheel!

SHORT A PIECE AT CLEAN-UP TIME?

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

✓ VOLV-01 4-POINT CLEAN-UP CHECKLIST

1. ALIGN CURSOR TO 0°:	Rotate the Gold cursor arm back to the 0° (2π) home position.
2. CHECK PIVOT CAP:	Ensure the center retention pin and cap are seated flush.
3. FOLD CARD TO [01]:	Fold this reference booklet to Panel [01] alongside the disc.
4. SLIDE INTO SLEEVE:	Slip the Volvelle disc and folded guide into the protective sleeve!

MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)
Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2). All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

SLIPCASE TRAVEL SLEEVE • KEEPS YOUR VOLVELLE SAFE IN ANY BINDER

NEXT: CONNECTED LAB ▼ PAGE 07

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

THE STACKCALC CONNECTED LAB

How Your Kits & RPN Calculator Stack Work Together

ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

WHERE YOUR BOX FITS IN THE 9-KIT LAB

BASE-10 **Ten-Frames Lab** Early Counting & Regrouping	DOMI-01 **Math Domino Studio** Doubles & 12-Frame Part-Whole	FRAC-01 **Fraction Studio** Proportional Towers & 4-Way Faces
PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers	ALGB-01 **Algebra Studio** Quadratic Area Arrays & Factoring	TIME-01 **Planetary Clock** 12:1 Geared Hands & Time Wedges
PROB-01 **Probability Dice** Fair Polyhedrals & 10-Sector Spinner	VOLV-01 ◀ YOUR BOX **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals	SC-32-HW **SC-32 RPN Calculator** 4-Register Stack Scientific Engine

WHY THIS MATTERS: ZERO "THROW-AWAY" MATH

In most classrooms, every grade uses a totally unrelated plastic toy. In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. VOLV-01 BOX LID ICON

DEG RAD 6

[DEG▸RAD] & [↱][▸y,x]
EMBOSSED ON BOX LID
Build First → Press Key!

2. SC-6 ELEMENTARY RPN

3. SC-32 SCIENTIFIC RPN

▲ BOX ICON ↔ CALCULATOR KEY CONTINUITY:
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

HOW BLOCKS CONNECT TO THE CALCULATOR

SHARED MODULAR FOOTPRINT & HEIGHTS

- Every 30° (π/6) step on the Volvelle matches one hour on the Classroom Planetary Clock!
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

WHY WE CALL IT "STACKCALC" (THE RPN SECRET)

- Pair with your SC-32 calculator: read exact radicals on the wheel, then compute decimals on the RPN stack.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 0 (1st Block)
X: 1 ($\cos^2 45° + \sin^2 45°$) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. PLACE 1ST PIECE ON STAGE:
Type number + [ENTER] → lifts to Y

2. ADD / STACK 2ND PIECE:
Type 2nd number in X, then [+]/[×]

3. PRESS BOX ICON KEY:
Press [DEG▸RAD] & [↱][▸y,x]
✓ **Zero Parentheses • Always Visible!**

BOX ICON = [DEG▸RAD] & [↱][▸y,x] • VERIFIED LESSON SEQUENCES:

- Lesson 1: 45 [COS] [↰] [x^2] 45 [SIN] [↰] [x^2] [+] → X = 1
- Lesson 2: 30 [ENTER] 1 [↱] [▸y,x] → X = 0.86602540

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

KEEP THIS CARD! PART IDs & VIDEO

Complete 5-Piece Reference & 60-Second 3D Video Guide

PART ID REFERENCE TABLE (5 TOTAL PIECES)

✓ KEEP THIS CARD INSIDE YOUR BOX LID!

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

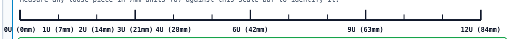
PART NAME	QTY	COLOR / FINISH	PART ID
Base Stator Disc (0°–360°)	1x	Black	VOLV-VOLV-a17
Polar Radian Rotor Disc	1x	Gold	VOLV-VOLV-a17
Exact Radical Cursor Arm	1x	Blue	VOLV-VOLV-a17
Snap-Fit Center Hub Pin	1x	Clear	VOLV-VOLV-a17
Protective Travel Slipcase	1x	Gray	VOLV-VOLV-a17

NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.
Enter any Part ID above at **stackcalc.io/parts** to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY — MATCH ANY LOOSE PART

Black	Gold	Blue	Gray
Base Stator Disc	Polar Radian Rotor	Radial Cursor Arm	Protective Slipcase

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)
Measure any loose piece in 7mm units (U) against this scale bar to identify it:



✓ DISTRICT PO & SAFETY: 10-Pack SKU VOLV-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO
stackcalc.io/products/unit-circle-volvelle/
See animated 3D examples & pack-up walkthrough

QUESTIONS, CLASSROOM SETS & SUPPORT

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

Designed in Mountain View, CA • Non-Toxic & Washable

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

BOOK CHAPTERS: SC-32 Manual Ch. 5 (Trigonometry & Phasors) • Appendix C
STANDARDS: CCSS HSF-TF.A.1, HSF-TF.A.2, HSF-TF.C.8 • TEKS P.4A, P.4B (Unit Circle)
WATERLINE SPEC: 3-Layer Rotating Disc Assembly + Center Retention Pin

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SC32):
Ex 1: 45 [COS] [↰] [x^2] 45 [SIN] [↰] [x^2] [+] → X=1
Ex 2: 30 [ENTER] 1 [↱] [▸y,x] → X=0.86602540
Ex 3: 3 [√x] 2 [÷] 30 [COS] → X=0.86602540

BOX ICON KEY: [DEG▸RAD] & [↱][▸y,x]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!